

Class 12 Biology: High-Yield Diagram & Mark Breakdown

Chapter 9: Biotechnology – Principles and Processes (NCERT Page-wise Guide)

Instructions for Students: Practice sketching vectors, PCR temperature steps, and bioreactors neatly with pencil. Ensure all restriction sites, antibiotic resistance markers, and flow steps are labelled in BLOCK letters.

Fig No.	Diagram / Flowchart Title	NCERT	Marks	Priority	Key Components / Labelling Points
Fig 9.4	<i>E. coli</i> Cloning Vector pBR322	Pg 169	3 – 5 M	Top Priority	Markers: amp ^R (Pvu I, Pst I), tet ^R (BamH I, Sal I); Sites: EcoR I, Cla I, Hind III; ori, rop (Pvu II)
Fig 9.6	Polymerase Chain Reaction (PCR)	Pg 172	3 – 5 M	Top Priority	3 Steps: (1) Denaturation (Heat), (2) Annealing (Primers), (3) Extension (Taq polymerase + dNTPs) → ~1 billion copies
Fig 9.7 (a, b)	Bioreactor Types (Simple vs Sparged Stirred-Tank)	Pg 174	3 M	Top Priority	Motor, Impeller blades, Foam breaker, pH/Temp control, Steam/Air ports, Sparger (bubbles for surface area)
Fig 9.1	Action of Restriction Enzyme (EcoRI) & Sticky Ends	Pg 166	3 M	Top Priority	Palindromic sequence 5'-GAATTC-3', staggered cuts between G & A, sticky ends, ligation by DNA ligase
Fig 9.2	Recombinant DNA Technology Process Flow	Pg 167	3 M	High Priority	Foreign DNA + Vector DNA → Cut with same RE → Ligase → rDNA → <i>E. coli</i> Host → Transformation & Division
Fig 9.3	Agarose Gel Electrophoresis Setup & Bands	Pg 168	2 – 3 M	High Priority	Wells (-ve cathode), Migration to +ve anode, Largest to smallest bands, Ethidium bromide + UV light, Elution
Fig 9.5	DNA Spooling	Pg 171	1 – 2 M	Moderate	Precipitation of purified DNA with chilled ethanol into fine threads; spooling collection

EXAM STRATEGY & GUIDELINES

- **pBR322 Vector Accuracy (3–5 Marks):** Restriction sites must be in their exact genes (BamH I/Sal I in *tet^R*; Pst I/Pvu I in *amp^R*; Pvu II in *rop*).
- **PCR Steps (3–5 Marks):** State the purpose of each step and highlight thermostable *Taq polymerase* isolated from *Thermus aquaticus*.
- **Bioreactors (3 Marks):** Emphasize how the sparged reactor's bubbles dramatically increase the surface area for oxygen transfer.